

CS6886 Assignment 2: MobileNet-v2 on CIFAR-10

Compression by pruning, quantization and Huffman coding

Repository: <https://github.com/preethibosco/CS6886-A2>

Summary

Accuracy without compression	95.17% top-1
Model size, fp32	8.662 MB (2,270,794 values)
Best model compression ratio	35.90x
Best weight compression ratio	35.09x
Best activation compression ratio	4.00x
Accuracy after compression	93.21% (-1.96)
Final model size	0.241 MB
Storage overheads	see Q2(c)
Wandb parallel coordinates	Figure 3

The reported configuration is 3-bit weights, 8-bit activations, 70% sparsity, BatchNorm folded into the convolutions.

Q1. Training baseline

(a) Data preparation

Normalisation uses the CIFAR-10 training-set channel statistics, mean (0.4914, 0.4822, 0.4465) and standard deviation (0.2470, 0.2435, 0.2616).

Train transforms, in order:

1. `RandomCrop(32, padding=4)`. Pad 4 px each side, then crop back to 32x32.
2. `RandomHorizontalFlip(p=0.5)`.
3. `ToTensor()`. HWC uint8 to CHW float in [0,1].
4. `Normalize(mean, std)`.
5. `RandomErasing(p=0.25, scale=(0.02, 0.20))`. Applied after normalisation, so the erased patch is filled at the channel mean.

Test transforms: `ToTensor()` and `Normalize` only.

A third split of 1024 images is drawn from the training set with the test-time transform. It is used only to calibrate activation ranges, so that no clipping threshold is ever fitted on test data.

(b) Model and training configuration

MobileNet-v2 is implemented from scratch rather than imported. The ImageNet configuration downsamples 32x, which on a 32x32 input reaches a 1x1 feature map by the c=64 stage. We remove two downsampling steps, the stem stride and the c=24 stage stride, both from 2 to 1. That gives 8x total downsampling and a 4x4 final feature map. Everything else follows the paper.

Setting	Value
Width multiplier	1.0
Dropout (before classifier)	0.2
BatchNorm momentum	0.05
Parameters	2,236,682 (8.532 MB fp32)
BatchNorm running buffers	34,112 (0.130 MB)
Optimiser	SGD, momentum 0.9, Nesterov
Learning rate	0.1, 5-epoch linear warmup then cosine to 0.0
Weight decay	0.0005, not applied to BatchNorm, biases or depthwise weights
Label smoothing	0.1
Epochs / batch size	300 / 128
Precision	bfloat16 autocast
Seed	42

Weight decay is withheld from the depthwise convolutions because each channel holds only 9 weights, so the same decay that is mild for a 960x160 pointwise convolution is a strong pull toward zero here and can remove channels entirely.

(c) Results

Final test top-1 is **95.17%** at epoch 293, against 99.46% on the training set, a gap of 4.29 points. Curves are in Figure 1.

In Figure 1 the test loss sits below the train loss for the whole run. This is not an error: the training loss includes label smoothing and is measured on augmented images, while the test loss is plain cross-entropy on clean ones. The accuracy panel shows the real generalisation gap.

Errors are concentrated in one class pair rather than spread across the ten classes:

Class	Top-1	Most confused with
cat	88.20%	dog (6.2%)
dog	91.60%	cat (4.9%)
bird	94.10%	deer (1.4%)

The eight rigid-body classes are all at or above 94%. cat and dog form a single reciprocal confusion pair; at 32x32 the texture cues that separate them are near the resolution limit. Per-class accuracy and the confusion matrix are in Figure 2.

Q2. Compression implementation

(a) Method

Four stages, all written from scratch. No quantization, pruning or compression library is used anywhere; `scripts/conformance.py` enforces this with an AST scan of the source tree.

1. Pruning (`src/compress/prune.py`). Fine-grained magnitude pruning with a single global threshold across the prunable layers, capped so no layer exceeds 95% sparsity. A global threshold lets the network decide where sparsity belongs instead of forcing every layer to the same ratio.

Surviving weights need their positions recorded. Two encodings are implemented and the cheaper one is chosen per layer:

- *bitmap*: one presence bit per weight plus codes for the survivors. Costs $N + \text{nnz} \cdot b$ bits.
- *relative index*: a delta to the previous survivor, with filler entries when a gap exceeds the largest representable delta. Costs $\text{entries} \cdot (b + \text{index_bits})$.

The delta width is searched per layer over 3, 4, 5, 6 and 8 bits rather than fixed at the 4 bits used in Deep Compression. A 4-bit delta spans 15 positions, so at 98% sparsity the mean gap exceeds it and filler entries outnumber real values roughly 3:1. Searching the width recovers 30% of the encoded size at 95% sparsity and 60% at 98%.

2. Weight quantization (`src/compress/weight_share.py`, `quantize.py`). Three methods are implemented and selectable: k-means weight sharing (Lloyd’s algorithm, linear centroid initialisation), linear quantization per output channel, and linear quantization per tensor. Symmetric quantization is used for weights throughout:

$$q = \text{clamp}(\text{round}(x/s), -2^{b-1} + 1, 2^{b-1} - 1), \quad s = \max |x| / (2^{b-1} - 1)$$

3. Huffman coding (`src/compress/huffman.py`). Canonical Huffman over the quantized codes and over the index deltas, coded separately. Canonical form means the table is one code length per symbol rather than a serialised tree. Entropy coding is applied only where it wins: on short streams the table exceeds the saving, and the raw stream is kept instead.

4. Activation quantization (`src/compress/quantize.py`). Ranges are observed on the calibration split as an exponential moving average of per-batch 99.99th percentiles, then frozen. The scheme differs by site: tensors after ReLU6 are one-sided in $[0,6]$ and use asymmetric quantization, so all 2^b codes fall in the range that occurs; tensors after a linear-bottleneck projection are signed and near zero-mean and use symmetric quantization.

Fine-tuning (`src/compress/qat.py`). One-shot compression at these settings is not usable. At 70% sparsity and 3-bit weights the model drops to 10.00%. Pruning masks are fixed and the model is retrained with quantization in the forward pass, using the straight-through estimator in its master-weight form: quantize in place, forward and backward at the quantized point, restore the full-precision master, then step. The mask is re-applied after each step, because momentum and weight decay both move pruned weights away from zero.

Gradient clipping (norm 5.0) is required, not optional. Without it the estimator diverges at low bit width: an instrumented run reached a gradient norm of $8.3e5$ by iteration 25, drove one layer’s weights to $6.7e14$, and the loss then stayed at $\ln(10)$ for the rest of training. A dead layer has no gradient, so the failure cannot recover, and whether it triggers depends on batch order.

(b) Application to MobileNet-v2

Layers	Parameters	Share	Treatment
Pointwise 1x1 convolutions	2,124,672	94.99%	pruned and quantized to 3 bits
Depthwise 3x3 convolutions	64,224	2.87%	quantized to 8 bits, not pruned
BatchNorm	68,224	3.00%	folded into the preceding convolution
Classifier	12,810	0.57%	quantized to 8 bits, not pruned
Stem convolution	864	0.04%	quantized to 8 bits, not pruned

The exceptions follow from the parameter distribution. Pointwise convolutions are 95% of the model, so compression has to happen there. Depthwise convolutions are 2.9%: each channel has 9 weights and no cross-channel mixing, so there is little redundancy to prune and no neighbouring channel to absorb quantization error. Exempting them costs almost nothing in ratio. The stem and classifier are 0.6% combined and sit at the input and output, where error is not attenuated by any later layer.

BatchNorm is folded into the preceding convolution rather than quantized. At inference BatchNorm is an affine map with frozen statistics, so

$$W' = W \cdot \gamma / \sqrt{\sigma^2 + \epsilon}, \quad b' = \beta - \gamma \mu / \sqrt{\sigma^2 + \epsilon}$$

is exact. Folding replaces four stored vectors per layer with one fused bias. The transform is verified by comparing outputs before and after: maximum logit difference 9.4e-13 over 52 folded pairs, with identical predictions.

(c) Storage overheads

Everything a decoder needs is charged. Sizes are computed in bits and converted to MB at the end.

Item	Included as
Quantized weight codes	Huffman-coded stream where that is cheaper, else fixed-width
Sparse position metadata	bitmap bits, or delta bits including filler entries
Quantization parameters	2 ^b fp32 centroids (k-means), or fp32 scales (linear)
Huffman code tables	5 bits per alphabet symbol, canonical form
BatchNorm	none, folded away
Biases	fp16

Two items are easy to miss and both are counted. `model.parameters()` does not return the BatchNorm `running_mean` and `running_var` buffers, but inference needs them: 34,112 values, 0.130 MB in fp32. They are included in the 8.662 MB baseline, so the compression ratio is not inflated by understating it. Second, the quantization metadata is not negligible after compression: a 256-entry fp32 codebook is 8,192 bits, more than the 6,912 bits of values in the 864-weight stem layer.

Reported sizes are not analytic estimates. Every encoder has a matching decoder, and `scripts/conformance.py` decodes the claimed bitstream and checks it reproduces the weights the accuracy number was measured with.

Q3. Compression results

(a) Compression levels

Two sweeps were run. A post-training sweep over 60 configurations (weight bits in {2,3,4,6,8}, activation bits in {2,4,6,8}, sparsity in {0, 0.5, 0.8}), and a fine-tuned sweep over 23 configurations covering the useful region.

(b) Accuracy

Figure 3 is the parallel coordinates chart over the post-training sweep, built from the same runs logged to `Weights & Biases`. Post-training quantization alone tops out near 15x before accuracy falls away; the useful range needs fine-tuning.

Fine-tuned results, Pareto front first (Figure 4 plots all of them):

w	a	sparsity	fold	ratio	size (MB)	PTQ	after fine-tuning
3	8	0.90	yes	48.90x	0.177	10.00%	83.28%
3	8	0.80	yes	38.67x	0.224	10.12%	92.77%
3	8	0.70	yes	35.90x	0.241	10.00%	93.21%
3	8	0.50	yes	32.66x	0.265	10.00%	93.59%
3	8	0.90	no	32.23x	0.269	15.41%	93.63%
3	8	0.70	no	31.59x	0.274	32.42%	94.20%
3	8	0.80	no	31.40x	0.276	27.12%	94.24%
4	8	0.80	no	23.31x	0.372	45.18%	94.73%

Two results follow. First, 3-bit weights at moderate sparsity beat 4-bit weights at heavy sparsity: w3/sp0.80 and w4/sp0.95 reach the same ratio, 31.4x against 31.3x, but differ by 1.77 points of top-1. 3-bit per-tensor quantization is already an implicit pruner: 88.3% of weights round to zero at sparsity 0. Explicit pruning therefore removes redundancy that has largely gone, while still paying for index metadata. Second, and for the same reason, raising sparsity from 70% to 90% at 3 bits moves the ratio only 31.59x to 32.23x and costs 0.6 points.

Q4. Compression analysis

Reported configuration: 3-bit weights, 8-bit activations, 70% sparsity, BatchNorm folded.

(a) **Weight compression ratio: 35.09x.** Conv and linear weights only, fp32 against the encoded bitstream including index metadata, quantization parameters and Huffman tables.

(b) **Activation compression ratio: 4.00x.** Measured by pushing one image through the network and recording the output tensor of every quantization site (52 sites: 35 after ReLU6, 17 after the residual add). The ratio is the sum over those tensors of 32 bits per element, divided by the sum of 8 bits per element, which is the reduction in activation traffic over one inference. Peak single-tensor activation, which is what bounds an on-chip buffer, is 147,456 elements: 0.562 MB fp32 against 0.141 MB quantized, the same ratio.

(c) **Accuracy at that ratio: 93.21%,** against 95.17% uncompressed (-1.96). The same configuration without fine-tuning gives 10.00%.

(d) **Final model size: 0.241 MB,** from 8.662 MB, a model compression ratio of **35.90x.**

Q5. Reproducibility

(a) Training, evaluation and compression are separate import paths. Nothing in `src/compress/` imports `src/train.py`, and `src/models/mobilenetv2.py` contains no compression code. The activation quantizers attach through forward hooks.

(b) `README.md` gives the exact commands, the environment and pinned dependency versions (torch 2.11.0+cu128, torchvision 0.26.0+cu128, numpy 2.2.6, wandb 0.29.0, Python 3.10). Seed defaults to 42 everywhere and is set through `src.utils.seed_everything`, which seeds Python, NumPy and Torch on CPU and CUDA. Compression and evaluation scripts additionally enable `cuda.deterministic` and set `CUBLAS_WORKSPACE_CONFIG`, so all reported compression numbers reproduce exactly. Baseline training leaves `cuda.benchmark` on for speed, so it is seed-controlled but not bit-deterministic.

(c) <https://github.com/preethibosco/CS6886-A2>

Figures

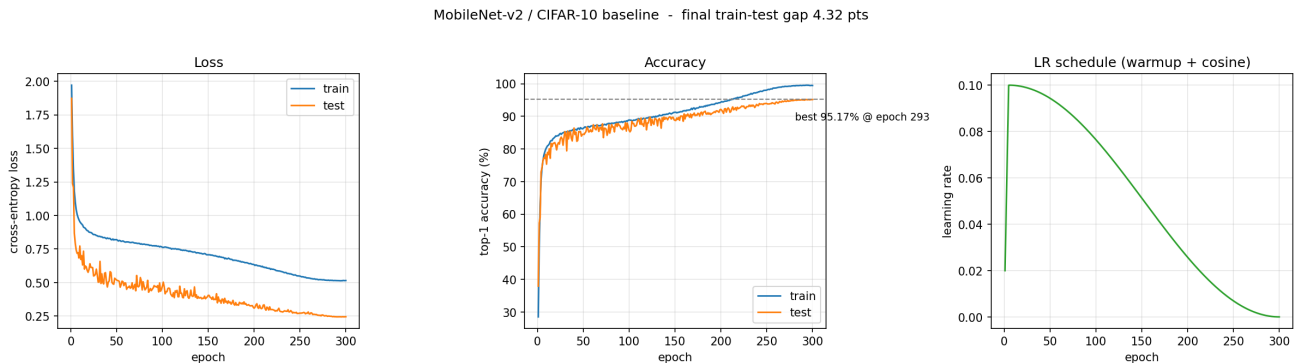


Figure 1: Baseline training. Test loss is below train loss because the training loss includes label smoothing and is measured on augmented images.

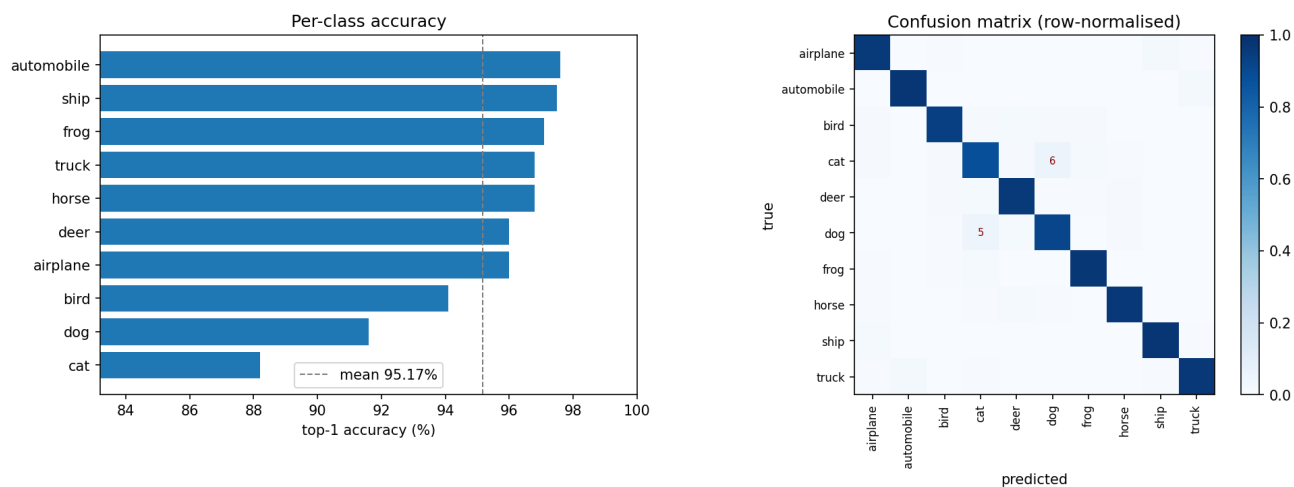


Figure 2: Per-class accuracy and confusion matrix for the uncompressed model.

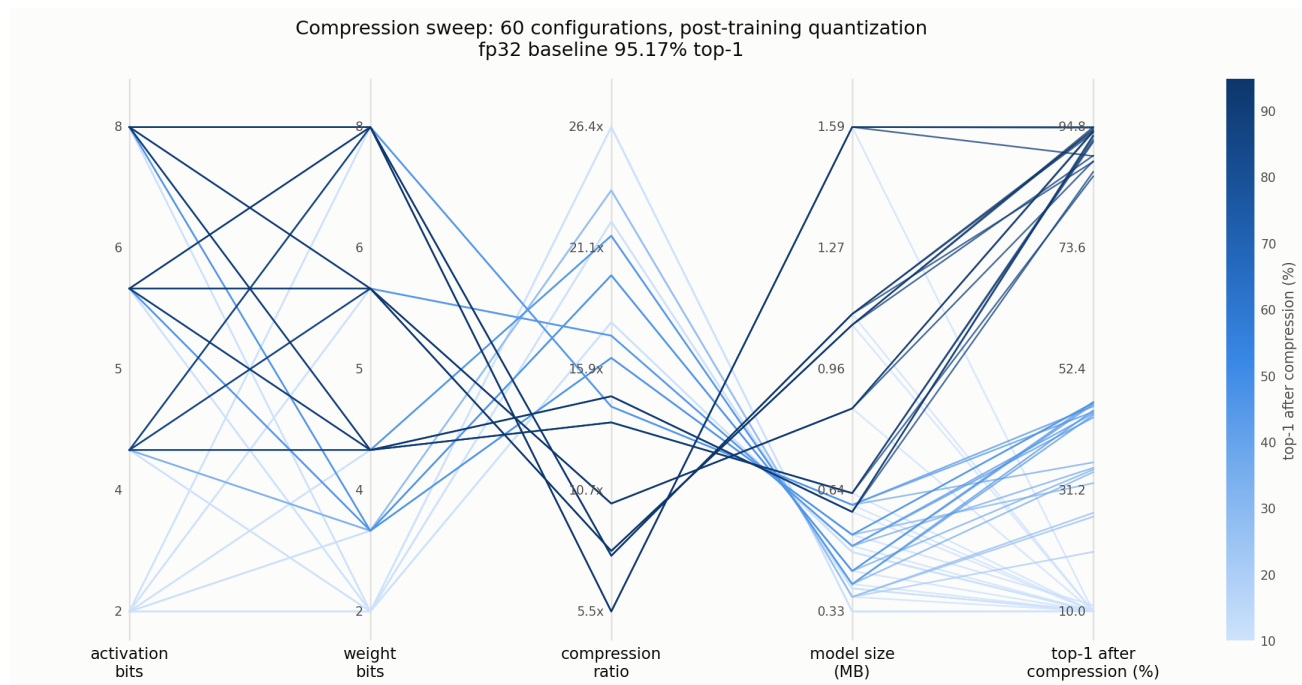


Figure 3: Parallel coordinates over the 60-configuration post-training sweep.

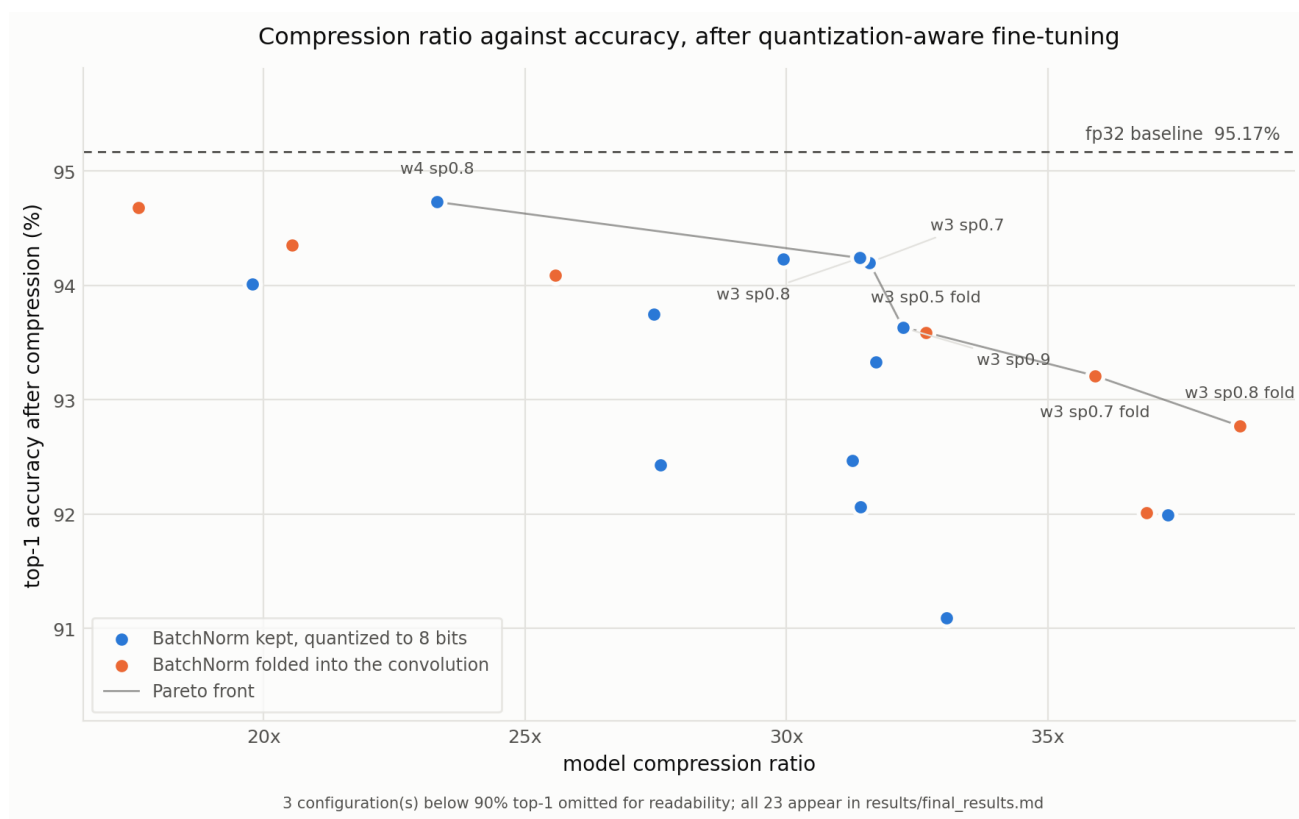


Figure 4: Compression ratio against accuracy after fine-tuning.